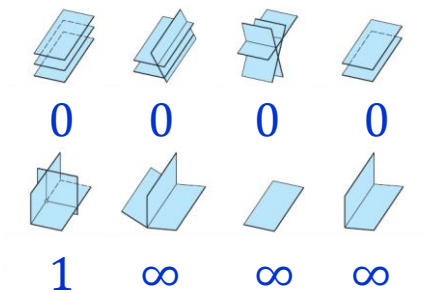


Subspaces

Video 9.1.1

When we are in low dimensions (such as systems with two variables x_1, x_2 or with three variables x_1, x_2, x_3) we can use our geometric intuition to understand linear algebra problems.

For example,
when we were
thinking about
numbers of
solutions in $\mathbb{R}^3$:



The reason we can do this is that *we actually live* in 3 dimensions and so our brains can understand solution sets in terms of familiar geometric objects we can see like *lines* and *planes*.

But what about when we are in higher dimensions, such as a system with 7 variables, $x_1, \dots, x_7$? Can we find some way of thinking geometrically in $\mathbb{R}^7$?

For example, can we talk about the “*dimension*” of a set of solutions in $\mathbb{R}^7$?

Yes we can but to do this we’ll need to think about what is special about lines and planes.

The starting point is to generalize the concepts “line through the origin” and “plane through the origin” to the abstract concept of a **subspace of $\mathbb{R}^n$** .

Subspaces of $\mathbb{R}^n$
are just certain
special subsets of
 $\mathbb{R}^n$ satisfying a list
of three properties.

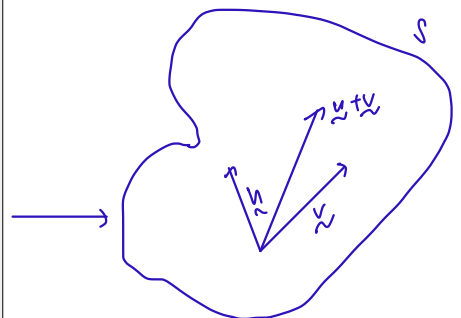
subset is very close to
subspace

Definition ('Subspace of $\mathbb{R}^n$ ').

Consider some subset S of some Euclidean space $\mathbb{R}^n$:
 $S \subset \mathbb{R}^n$.

Then S is said to be a **subspace of $\mathbb{R}^n$** if it satisfies the following three properties:

1. (**Subspace axiom 1:** Non-empty.) The subset S is **non-empty**.
2. (**Subspace axiom 2:** Closure under addition.) Whenever $\vec{u}$ and $\vec{v}$ are vectors in S then the vector $\vec{u} + \vec{v}$ also lies in S .
3. (**Subspace axiom 3:** Closure under scalar multiplication.) Whenever $\vec{u}$ is a vector in S , and $c \in \mathbb{R}$ is any scalar, then the vector $c\vec{u}$ also lies in S .



Anything you build from vectors that are inside S is gonna stay in S .

In the following slides we'll intuitively discuss the following subsets of $\mathbb{R}^2$.

Exercise:

Are any of the following subsets of $\mathbb{R}^2$ actually subspaces?

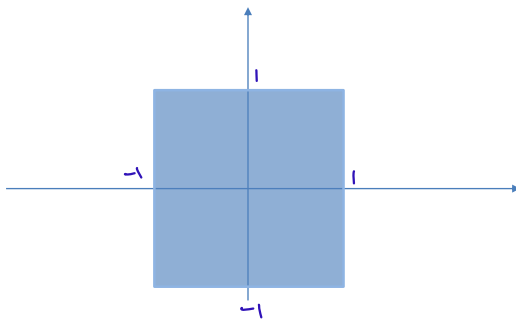
- $S = \{(x_1, x_2) \in \mathbb{R}^2; -1 \leq x_1 \leq 1, -1 \leq x_2 \leq 1\}$
- $S = \{(x_1, x_2) \in \mathbb{R}^2; x_1 \geq 0\}$
- $S = \{(x_1, x_2) \in \mathbb{R}^2; x_1 + x_2 = 0\}$
- $S = \{(x_1, x_2) \in \mathbb{R}^2; x_1 + x_2 = 1\}$
- $S = \{(0,0)\}$.

Before doing some careful algebra, let's test our intuition for this definition with some low dimensional examples.

Video 9.1.2

Example

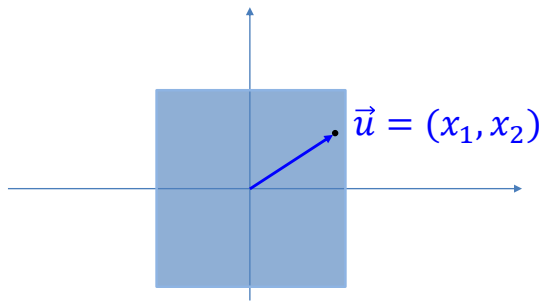
Is the following subset of $\mathbb{R}^2$ a subspace of $\mathbb{R}^2$?



$$S = \{(x_1, x_2) \in \mathbb{R}^2; -1 \leq x_1 \leq 1, -1 \leq x_2 \leq 1\}$$

Quick reminder:

Here we are showing a subset of points in $\mathbb{R}^2$. If we want to think of these vectors as arrows (for example so we can think geometrically about operations) then a point corresponds to the arrow from the origin to that point.



$$S = \{(x_1, x_2) \in \mathbb{R}^2; -1 \leq x_1 \leq 1, -1 \leq x_2 \leq 1\}$$

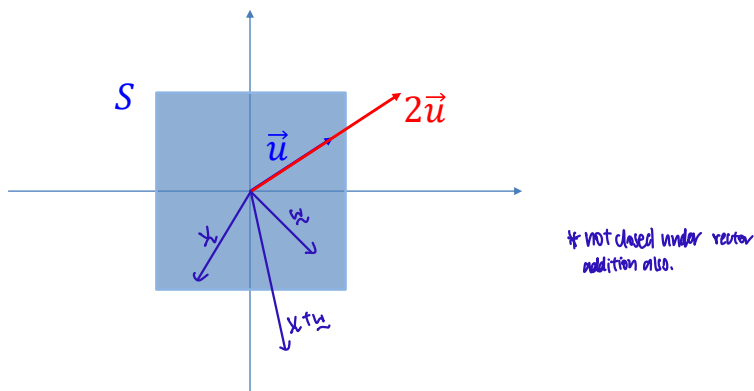
Discussion

No this subset is not a subspace.

For example it isn't closed under scalar multiplication.

Consider the vector $\vec{u} \in S$ shown below.

The vector $2\vec{u}$ no longer lies in S .

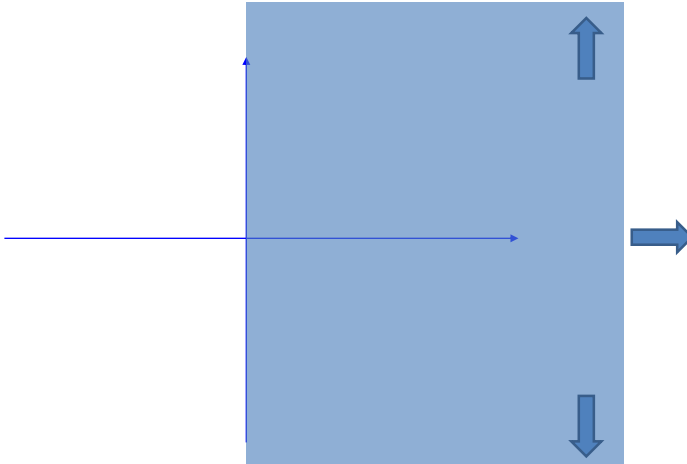


Another Example

Video 9.1.3

Is the following subset of $\mathbb{R}^2$ a subspace of $\mathbb{R}^2$? Which axioms hold?

$$S = \{(x_1, x_2) \in \mathbb{R}^2; x_1 \geq 0\}$$

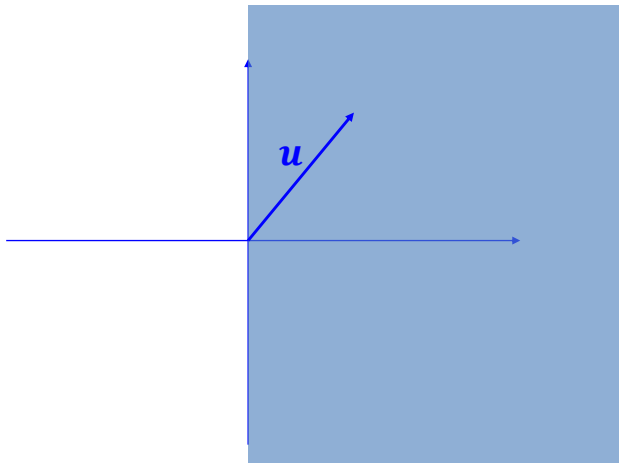


Discussion

Let's check the three axioms of a subspace.

Axiom 1: Is it non-empty?

Yes, obviously this subset contains some vectors.



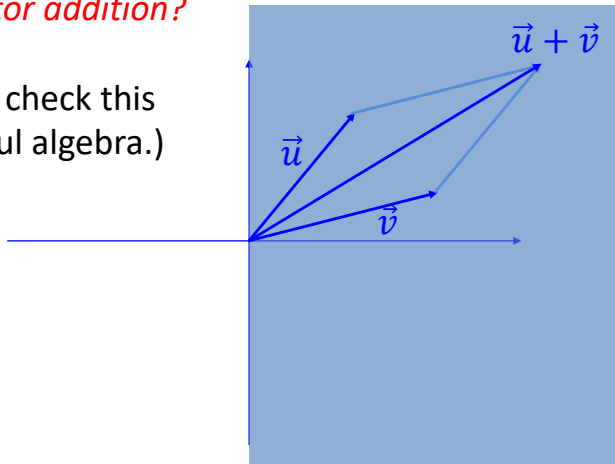
Discussion

Let's check the three axioms of a subspace.

Axiom 2: Is it closed under vector addition?

Yes, it is.

(We could check this with careful algebra.)



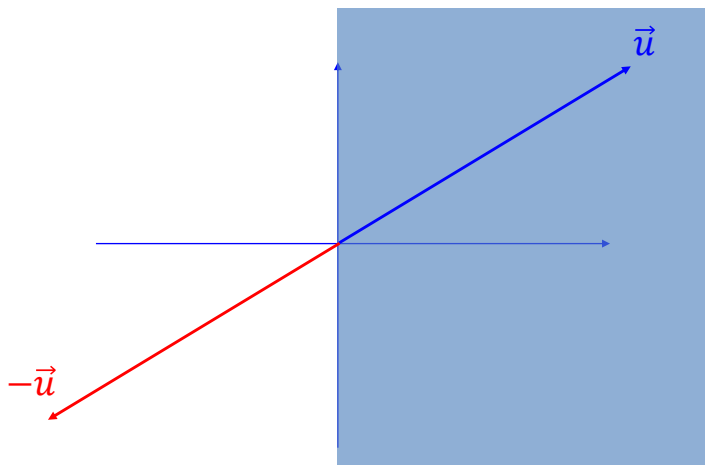
Sum of two vectors whose first coordinate is ≥ 0 is a vector such that the first coordinate is also ≥ 0 .

Discussion

Let's check the three axioms of a subspace.

Axiom 3: Is it closed under scalar multiplication?

NO, it isn't.



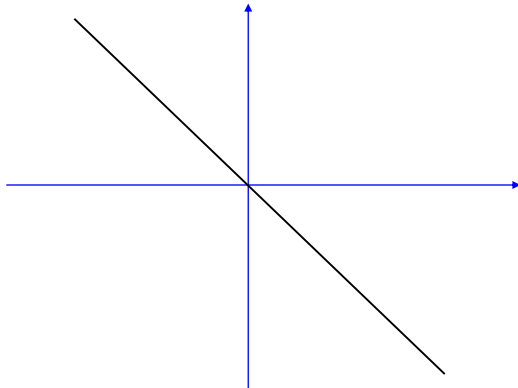
if positive scalar, it would be in the subset.

This subset is still not a subspace.

Example

Video 9.1.4

Is the following subset of $\mathbb{R}^2$ a subspace of $\mathbb{R}^2$?



$$\begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{matrix} \swarrow \\ x_1 \end{matrix} + \begin{matrix} \searrow \\ x_2 \end{matrix} = 0$$

$$S = \{(x_1, x_2) \in \mathbb{R}^2; x_1 + x_2 = 0\}$$

Let's check the three axioms of a subspace.

Axiom 1: Is it non-empty?

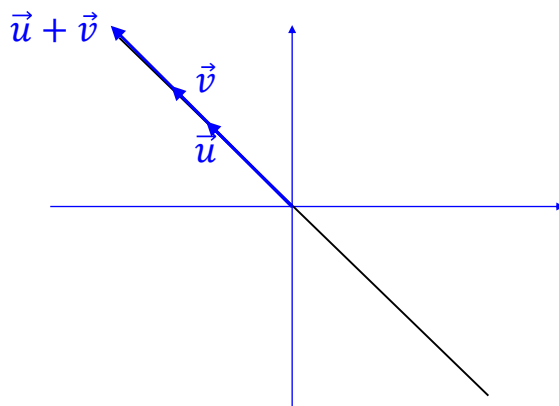
Yes, obviously this subset contains some vectors.

Example

Is the following subset of $\mathbb{R}^2$ a subspace of $\mathbb{R}^2$?

Axiom 2: Is it closed under addition?

Yes, it is. (We will check this later with some careful algebra.)



$$S = \{(x_1, x_2) \in \mathbb{R}^2, x_1 + x_2 = 0\}$$

any vector that lies in this subspace are all parallel.

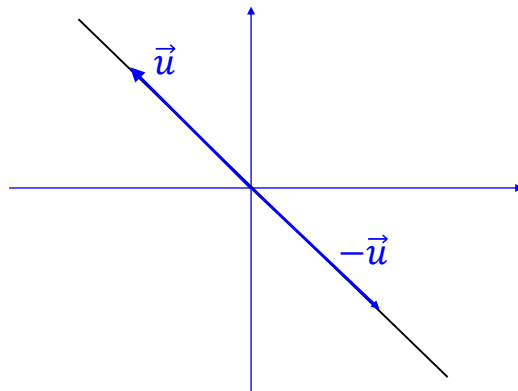
→ adding them up will give another vector parallel to the two vectors we started with.

Example

Is the following subset of $\mathbb{R}^2$ a subspace of $\mathbb{R}^2$?

Axiom 3: Is it closed under scalar multiplication?

Yes, it is. (Again, we must check this later with some careful algebra.)



$$S = \{(x_1, x_2) \in \mathbb{R}^2; x_1 + x_2 = 0\}$$

Stretching it still gets a vector lying on the line.

lines through origin is a subspace.

Example

Summary: The subset $S = \{(x_1, x_2) \in \mathbb{R}^2; x_1 + x_2 = 0\}$

- Is non-empty
- Is closed under vector addition
- Is closed under scalar multiplication.

So S is not just a subset of $\mathbb{R}^2$ but is a *subspace* of $\mathbb{R}^2$.

Example

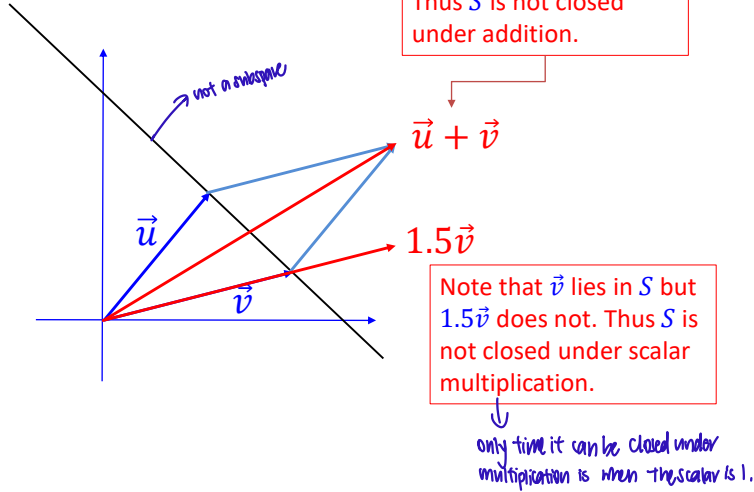
Video 9.1.5

Is the following subset of $\mathbb{R}^2$ a subspace of $\mathbb{R}^2$?

No. It is definitely non-empty so it satisfies axiom 1, but it fails the other two axioms.

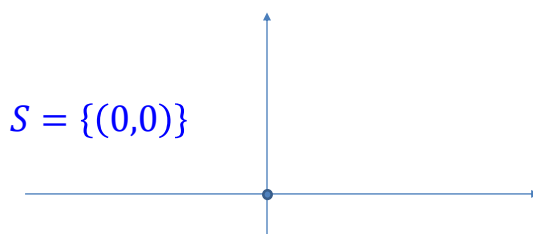
* To be a subspace, the subset has to go through the origin.

$$S = \{(x_1, x_2) \in \mathbb{R}^2; x_1 + x_2 = 1\}$$

Example

Video 9.1.6

Is the following subset of $\mathbb{R}^2$ a subspace of $\mathbb{R}^2$?



If it is another point, it is not a subspace.

Yes.

- It is non-empty.
- It is closed under vector addition: $(0,0) + (0,0) = (0 + 0, 0 + 0) = (0,0)$.
- It is closed under scalar multiplication: $c(0,0) = (c \cdot 0, c \cdot 0) = (0,0)$.

Example

What about all of $\mathbb{R}^2$?

Yes.

According to our definition it is actually a subspace of $\mathbb{R}^2$.

A proper subspace is a subspace different to the whole set.

(Recall our definition of *subset* includes the possibility of being every element in the original set.)

- It is non-empty.
- It is obviously closed under vector addition.
- It is obviously closed under scalar multiplication.

Question

Video 9.1.7

Can you think of an example of a subset $S \subset \mathbb{R}^2$ such that

- S is non-empty.
- S is closed under scalar multiplication.
- But S is not closed under vector addition?

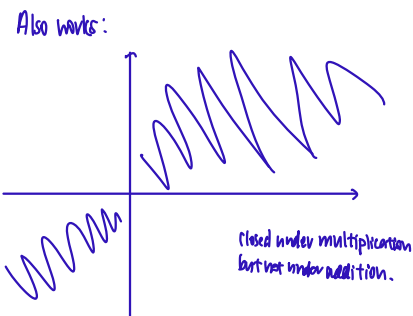
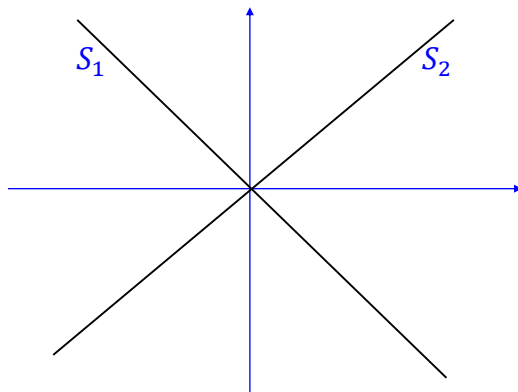
Discussion

Here's one example. Consider the following subsets of $\mathbb{R}^2$:

- $S_1 = \{(x_1, x_2) \in \mathbb{R}^2; x_1 + x_2 = 0\}$
- $S_2 = \{(x_1, x_2) \in \mathbb{R}^2; x_1 - x_2 = 0\}$.

Let S be the union of these two subsets:

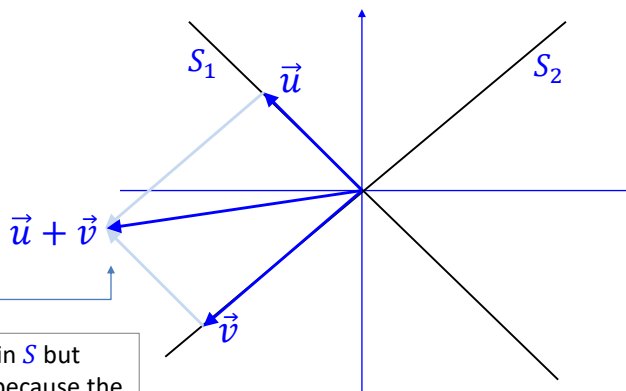
$$S = S_1 \cup S_2 \subset \mathbb{R}^2. \rightarrow \text{points in first line as well as points in second line}$$



Discussion

You can see this S is non-empty, and closed under scalar multiplication, but is not closed under vector addition.

it's a line



Note that $\vec{u}$ and $\vec{v}$ lie in S but that $\vec{u} + \vec{v}$ does not, because the tip of this arrow from the origin doesn't lie in the black lines.

Let's summarize the geometric possibilities in the familiar low dimensions as suggested by our intuitive experiments:

Video 9.1.8

In dimension 2:

The *subspaces* of $\mathbb{R}^2$ are exactly the following possibilities:

1. The set containing just the origin $\{(0,0)\} \subset \mathbb{R}^2$. $\hookrightarrow$ dimension 0
 $\downarrow$
it contains just the origin
2. Lines containing the origin. $\rightarrow$ dimension 1
3. All of $\mathbb{R}^2$. $\rightarrow$ dimension 2

In dimension 3:

The *subspaces* of $\mathbb{R}^3$ are exactly the following possibilities:

1. The set containing just the origin $\{(0,0,0)\} \subset \mathbb{R}^3$. $\hookrightarrow$ dimension 0
2. Lines containing the origin. dimension 1
3. Planes containing the origin. dimension 2
4. All of $\mathbb{R}^3$. dimension 3

We can prove this is a correct list of subspaces in $\mathbb{R}^2$ and $\mathbb{R}^3$ later on.

To do that we need to first develop a theory of *the dimension of a subspace*.

This will be the main big topic discussed in the remainder of this course.

Discussion:

Let's pause and think about what we have just done.

We have found an abstract geometric property (the three subspace axioms) which in low dimensions specializes to the intuitive objects we are familiar with (lines through the origin, and planes through the origin).

But this abstract property is simple and something we can easily apply to subsets of higher dimensional Euclidean spaces $\mathbb{R}^n$.

Thus we have successfully found a way to generalise these familiar concepts to higher dimensions.

We will see that this formulation is very powerful, and there is a rich theory of "subspaces". Much of the intuition we hold about lines and planes carries over to higher dimension.

Using the axioms of a subspace carefully.

Video 9.2.1

So far we have just been thinking about subspaces intuitively.

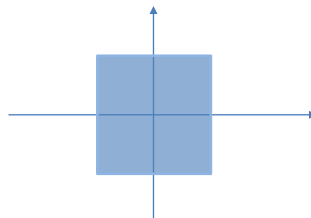
But to **build** strong mathematics we need to eventually work rigorously with the axioms.
(Even though intuition is always important.)

To start we'll look again at these examples we just discussed and see how to carefully prove with certainty that they are or are not subspaces.

Example

Is the following subset of $\mathbb{R}^2$ a subspace of $\mathbb{R}^2$?

$$S = \{(x_1, x_2) \in \mathbb{R}^2; -1 \leq x_1 \leq 1, -1 \leq x_2 \leq 1\}$$



Recall a general principle of the logic when discussing axioms:

- To prove something satisfies a rule you have to explain why the rule is satisfied *for every possible example*.
- But to show something doesn't satisfy an axiom, *one counter-example* is enough.

Example

Is the following subset of $\mathbb{R}^2$ a subspace of $\mathbb{R}^2$?

$$S = \{(x_1, x_2) \in \mathbb{R}^2; -1 \leq x_1 \leq 1, -1 \leq x_2 \leq 1\}$$

Discussion

In this case we want to show that this set is not a subspace. So it is enough to find *one counter-example to one of the axioms*.

We'll show a counter-example to the axiom about scalar multiplication. We have freedom to choose anything that demonstrates the axiom fails.

For example, we can choose $\vec{u} = (0.75, 0.75)$ and $c = 2$. $\rightarrow$ one single example, counter example.

Note that $-1 \leq 0.75 \leq 1$ so that this $\vec{u}$ satisfies the rule which defines this subset. Thus $\vec{u} \in S$.

Now if S was a subspace it would be closed under scalar multiplication. This would imply

$$c\vec{u} = 2(0.75, 0.75) = (1.5, 1.5) \in S.$$

But $1.5 \not\leq 1$ so $c\vec{u} \notin S$.

So we have demonstrated that S is not a subspace because it is not closed under scalar multiplication.

Another Example

Video 9.2.2

Use the axioms to carefully prove or disprove whether the following subset of $\mathbb{R}^2$

$$S = \{(x_1, x_2) \in \mathbb{R}^2, x_1 \geq 0\}$$

is a subspace of $\mathbb{R}^2$?

Solution

We will show that S is not closed under scalar multiplication, so that it is not a subspace.

Consider the vector $\vec{u} \in (1, 0)$. Note that $1 \geq 0$ so this vector satisfies the condition that defines S . Thus $\vec{u} \in S$.

If S was a subspace then it would be closed under scalar multiplication. So we would need that the result of scalar multiplication by (-1) , namely $(-1)(1, 0) = (-1, 0)$ lies again in S .

counter example

But $-1 \not\geq 0$ so $(-1)\vec{u} \notin S$. Thus S is not a subspace.

Now let's focus on an example that actually *is* a subspace.

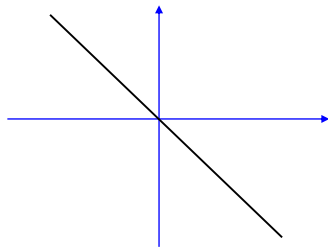
Video 9.2.3

Example

Is the following subset of $\mathbb{R}^2$

$$S = \{(x_1, x_2) \in \mathbb{R}^2, x_1 + x_2 = 0\}$$

a subspace of $\mathbb{R}^2$?



Note: This is a line through the origin so we expect this will be a subspace.

Example

Is the following subset of $\mathbb{R}^2$

$$S = \{(x_1, x_2) \in \mathbb{R}^2, x_1 + x_2 = 0\}$$

a subspace of $\mathbb{R}^2$? Prove it.

Observe: This is the important data. This is the rule which tells you which elements of $\mathbb{R}^2$ lie in S .

Discussion

This will be a case where we are proving the subset is actually a subspace.

So it isn't enough just to check specific examples.

We have to explain using logic and equations why the properties of a subspace hold *for every possible example of scalar multiplication and vector addition.*

Example

Is the following subset of $\mathbb{R}^2$

$$S = \{(x_1, x_2) \in \mathbb{R}^2, x_1 + x_2 = 0\}$$

a subspace of $\mathbb{R}^2$? Prove your answer.

↑
(*)

Proof

We'll go through the axioms in turn.

Axiom 1: How to prove S non-empty?

To show a set is non-empty we just need to find a vector in it. We usually consider the zero vector $\vec{0} = (0,0)$.

To see $\vec{0} \in S$ we check it satisfies the rule (*) that defines S .

Indeed:

(First entry of $\vec{0}$) + (Second entry of $\vec{0}$) = $0 + 0 = 0$.

Thus $\vec{0} \in S$ and S is non-empty as required.

Proof (continued).

Axiom 2: Why is S closed under vector addition?

What exactly do we need to prove?

For every possible pair of vectors $\vec{u} \in S$ and $\vec{v} \in S$ the sum $\vec{u} + \vec{v}$ again lies in S .

We start by writing: "Let $\vec{u}$ and $\vec{v}$ be arbitrary vectors from S ."

The word "arbitrary" means they can be any vector at all from S .

The **only** thing we know about the vectors is that they satisfy the rule (*) which defines S .

We'll need to use that in our explanation.

Let $\vec{u}$ and $\vec{v}$ be arbitrary vectors from S .

What exactly do we know about $\vec{u}$ and $\vec{v}$?

We know that

- $\vec{u} \in S$. So $\vec{u} = (u_1, u_2)$ where $u_1 + u_2 = 0$ (1).
- $\vec{v} \in S$. So $\vec{v} = (v_1, v_2)$ where $v_1 + v_2 = 0$ (2).

Now do a calculation to check the axiom:

The sum of these two vectors is:

$$\vec{u} + \vec{v} = (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2).$$

Now check this vector is again in S by checking it satisfies the equation defining S :

$$\begin{aligned} & \text{(1st entry of } (\vec{u} + \vec{v}) \text{) + (2nd entry of } (\vec{u} + \vec{v})) \\ = & (u_1 + v_1) + (u_2 + v_2) \\ = & (u_1 + u_2) + (v_1 + v_2) && \text{(Just rearranging)} \\ = & 0 + 0 && \text{(By (1) and (2))} \\ = & 0 \end{aligned}$$

Proof (continued).

Axiom 3: Why is S closed under scalar multiplication?

What exactly do we need to prove?

For every possible vector $\vec{u} \in S$ and every possible scalar $c \in \mathbb{R}$ the result of scalar multiplication lies back in S .

Proof (continued).

Axiom 3: Why is S closed under scalar multiplication?

Let $\vec{u} \in S$ be an arbitrary vector from S .

Let $c \in \mathbb{R}$ be an arbitrary scalar.

Because $\vec{u} \in S$ we know that $\vec{u} = (u_1, u_2)$ where $u_1 + u_2 = 0$.
Denote this equation (1).

We must check $c\vec{u} \in S$.

To check this we check that $c\vec{u} = (cu_1, cu_2)$ satisfies the equation that defines S :

$$\begin{aligned}
 & \text{(1st entry of } c\vec{u}) + \text{(2nd entry of } c\vec{u}) \\
 = & \quad cu_1 + cu_2 \\
 = & \quad c(u_1 + u_2) && \text{(Just rearranging)} \\
 = & \quad c * 0 && \text{(By (1).)} \\
 = & \quad 0
 \end{aligned}$$

Thus $c\vec{u} \in S$ and S is closed under scalar multiplication.

Proof (Conclusion).

We have proved that S satisfies all three axioms of a subspace.

Thus we have proved that S is a subspace.



Exercise

Video 9.2.4

Consider the following subset of $\mathbb{R}^3$

$$S = \{(x_1, x_2, x_3) \in \mathbb{R}^3, x_1 + 2x_2 - 3x_3 = 0\}.$$

Is it a subspace of $\mathbb{R}^3$?

Prove your answer.

requirement for a vector
to lie inside S

Proof

We will prove it is indeed a subspace.

• Note $\mathbb{R}^3$ w/ (w/o) have 2 free parameters.

• Or check $(0,0,0)$.

$$0 + 2(0) - 3(0) = 0$$

$$\therefore \vec{0} \in S$$

$\therefore S$ is non-empty.

Axiom 1 (Set is non-empty.)Axiom 1 (Set is non-empty.)The first thing we need to do is show that S is non-empty.To do this we'll show the zero vector $\vec{0} = (0,0,0)$ lies in S .And to do that just check the rule that defines S is satisfied by $\vec{0}$.

$$\begin{aligned} & (\text{1st entry of } \vec{0}) + 2 * (\text{2nd entry of } \vec{0}) - 3 * (\text{3rd entry of } \vec{0}) \\ = & 0 + 2 * 0 - 3 * 0 \\ = & 0. \end{aligned}$$

Thus $\vec{0}$ satisfies the rule which defines S and so $\vec{0} \in S$.

Axiom 2 (Closure under vector addition.)

Axiom 2 (Closure under vector addition.)

Let $\vec{u}$ and $\vec{v}$ be arbitrary vectors from S .

Thus $\vec{u} = (u_1, u_2, u_3)$ with $u_1 - 2u_2 + 3u_3 = 0$ (*).

Also $\vec{v} = (v_1, v_2, v_3)$ with $v_1 - 2v_2 + 3v_3 = 0$ (**).

Now we need to check that

$$\vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2, u_3 + v_3)$$

also lies in S . To do that we just check the rule that defines S is satisfied by this vector.

$$\begin{aligned} & (1\text{st entry}) + 2 * (2\text{nd entry}) - 3 * (3\text{rd entry}) \\ = & (u_1 + v_1) + 2(u_2 + v_2) - 3 * (u_3 + v_3) \\ = & (u_1 + 2u_2 - 3u_3) + (v_1 + 2v_2 - 3v_3) && (\text{Rearranging.}) \\ = & 0 + 0 && (\text{By (*) and (**).}) \\ = & 0 \end{aligned}$$

Thus $\vec{u} + \vec{v}$ satisfies the rule which defines S and so $\vec{u} + \vec{v} \in S$.

Take $\vec{u} \in S$. $\vec{u} = (u_1, u_2, u_3)$

$$u_1 + 2u_2 - 3u_3 = 0 \quad \text{--- (1)}$$

And $\vec{v} \in S$. $\vec{v} = (v_1, v_2, v_3)$

$$v_1 + 2v_2 - 3v_3 = 0 \quad \text{--- (2)}$$

What I need to do
show $\vec{u} + \vec{v} \in S$ by showing $\vec{u} + \vec{v}$ satisfies (1).

$\vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2, u_3 + v_3)$. Does it satisfy (1)?

$$\begin{aligned} \text{Ans is } & x_1 + 2x_2 - 3x_3 = 0 \\ \text{Let } & \vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2, u_3 + v_3) \\ & = (u_1 + 2u_2 - 3u_3) + (v_1 + 2v_2 - 3v_3) \\ & = 0 + 0 \\ & = 0 \end{aligned}$$

So $\vec{u} + \vec{v}$ satisfies (1).

So $\vec{u} + \vec{v} \in S$.

S is closed under vector addition.

Axiom 3 (Closure under scalar multiplication.)

Axiom 3 (Closure under scalar multiplication.)

Let $\vec{u}$ be an arbitrary vector from S and let $c \in \mathbb{R}$ be an arbitrary scalar.

Thus $\vec{u} = (u_1, u_2, u_3)$ with $u_1 - 2u_2 + 3u_3 = 0$ (*).

Now we need to check that

$$c\vec{u} = (cu_1, cu_2, cu_3)$$

also lies in S . To do that just check the rule that defines S is satisfied by this vector.

$$\begin{aligned} & (1\text{st entry}) + 2 * (2\text{nd entry}) - 3 * (3\text{rd entry}) \\ = & (cu_1) + 2(cu_2) - 3(cu_3) \\ = & c(u_1 + 2u_2 - 3u_3) && (\text{Rearranging.}) \\ = & c * 0 && (\text{By (*).}) \\ = & 0 \end{aligned}$$

Thus $c\vec{u}$ satisfies the rule which defines S and so $c\vec{u} \in S$.

Proof (Conclusion).

We have proved that S satisfies all three axioms of a subspace.

Thus we have proved that S is a subspace.



First properties of subspaces

Video 9.2.5

OK! So we have just defined a new concept (*subspace*) and we have explored a few preliminary examples. We introduced it for a reason: we wanted to find a way of formulating the idea of a line or a plane in a way which would make sense no matter what the dimension was.

Next: We'll start exploring what are the consequences of the set of rules (axioms) we have chosen for this concept.

As mathematicians we have a few questions to think about as we explore:

- Does this set of rules do what we want?
- Is there a rich, useful theory based on these rules?
- What are the properties of objects satisfying these rules?

The zero vector

Theorem

Let S be a subspace of $\mathbb{R}^n$.

Recall that the zero vector is the vector $\vec{0} = (0, \dots, 0) \in \mathbb{R}^n$.

The zero vector lies in S .

Our job is to explain why $\vec{0} \in S$ using only what we know about the subset $S \subset \mathbb{R}^n$: namely that it is a subspace, and so the three axioms of a subspace are true for S .

Proof.

- Because S is a subspace, by **Axiom 1**, it contains at least one vector $\vec{v} = (a_1, \dots, a_n)$.
- Also, because S is a subspace, by **Axiom 3**, it is closed under scalar multiplication so it also contains the vector $(0)\vec{v} = (0)(a_1, \dots, a_n) = (0, \dots, 0)$.
- Thus $\vec{0} \in S$.



Theorem (An equivalent set of axioms for subspace.)

Consider some subset S of some Euclidean space $\mathbb{R}^n$:

$$S \subset \mathbb{R}^n.$$

Then S is a **subspace of $\mathbb{R}^n$** (according to the definition we gave earlier) if and only if it satisfies the following three alternative axioms:

1. (*Alternative subspace axiom 1:* Contains the zero vector.) The subset S contains the zero vector: $\vec{0} \in S$.
2. (*Subspace axiom 2:* Closure under addition.) Whenever $\vec{u}$ and $\vec{v}$ are vectors in S then the vector $\vec{u} + \vec{v}$ also lies in S .
3. (*Subspace axiom 3:* Closure under scalar multiplication.) Whenever $\vec{u}$ is a vector in S , and $c \in \mathbb{R}$ is any scalar, then the vector $c\vec{u}$ also lies in S .

Intersections?

Video 9.2.6

We are currently starting to think about and explore our subspace axioms.

Subspaces are supposed to generalize “lines through the origin” and “planes through the origin” etc. One way of exploring these axioms would be to investigate whether properties of these low-dimensional objects still hold for general subspaces.

So what are some properties we can check?

One property those low dimensional objects have is that *when we take the intersection of two of them, we always get another one.*

For example, have a look at the list again.

In dimension 2:

The subspaces of $\mathbb{R}^2$ are exactly the following possibilities:

1. The set containing just the origin $\{(0,0)\} \subset \mathbb{R}^2$.
2. Lines containing the origin.
3. All of $\mathbb{R}^2$.

In dimension 3:

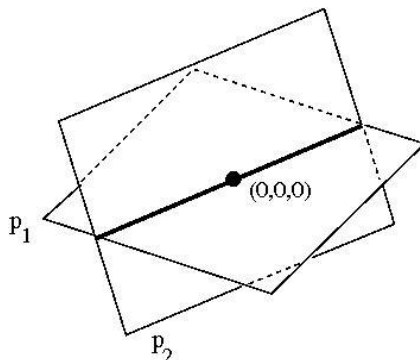
The subspaces of $\mathbb{R}^3$ are exactly the following possibilities:

1. The set containing just the origin $\{(0,0,0)\} \subset \mathbb{R}^3$.
2. Lines containing the origin.
3. Planes containing the origin.
4. All of $\mathbb{R}^3$.

Note that the intersection of any two objects on this list is always another object on the list.

Note that the intersection of any two objects on this list is always another object on the list.

For example, the intersection of two planes through the origin, is always a line through the origin.



You can check some other examples.

Indeed: it turns out that general subspaces have this property as well. (In fact the definition *simplifies* the story because you don't have to check lots of different types of pairs. One simple conceptual explanation works for all cases.)

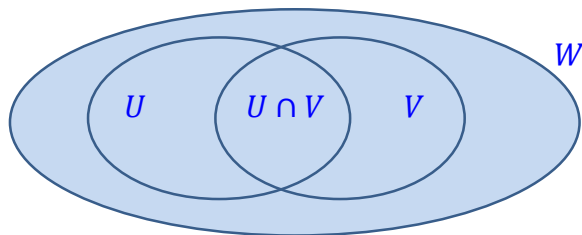
Theorem (The intersection of two subspaces is a subspace.)

Let S_1 and S_2 be subspaces of the same $\mathbb{R}^n$.

Then their intersection $S_1 \cap S_2$ is *also* a subspace.

Quick reminder about intersections:

If you have two subsets U and V of a larger set W , so $U \subset W$ and $V \subset W$ then the intersection $U \cap V$ is defined to be the set of elements of W that are in both U and V .



How to prove things about intersections?

If you need to prove some element x lies in $U \cap V$ then it is enough to

1. Prove $x \in U$.
2. and also prove $x \in V$.

Proof.

First let's organize our thoughts.

What information are we assuming?

We have a subset $S_1 \subset \mathbb{R}^n$ and we know that it is a subspace of $\mathbb{R}^n$ so that it satisfies three properties:

1. Contains the zero vector.
2. Closed under vector addition.
3. Closed under scalar multiplication.

We also know that we have a second subset $S_2 \subset \mathbb{R}^n$ which is also a subspace of $\mathbb{R}^n$ so it also satisfies the same three properties.

What do we need to show?

We need to show that $S_1 \cap S_2$ is a subspace. To do that we need to show that $S_1 \cap S_2$ satisfies the three axioms of a subspace.

Proof (continued).

We'll go through the axioms one-by-one.

Why does $S_1 \cap S_2$ contain the zero vector?

The intersection $S_1 \cap S_2$ is defined to be the set of vectors which lie both in S_1 and also in S_2 .

Now because we know that S_1 is a subspace, we know that it contains the zero vector $\vec{0} \in S_1$.

In exactly the same way we know that because S_2 is a subspace, it also contains the zero vector $\vec{0} \in S_2$.

So because $\vec{0} \in S_1$ and also $\vec{0} \in S_2$, also $\vec{0} \in S_1 \cap S_2$.

Why is $S_1 \cap S_2$ closed under vector addition?

Take two vectors $\vec{u}$ and $\vec{v}$ from $S_1 \cap S_2$.

We need to explain why $\vec{u} + \vec{v}$ is also in $S_1 \cap S_2$. We'll split that into three steps:

1. We'll explain that $\vec{u} + \vec{v} \in S_1$.
2. Then we'll explain that $\vec{u} + \vec{v} \in S_2$.
3. Thus $\vec{u} + \vec{v} \in S_1 \cap S_2$.

Step 1: Because $\vec{u} \in S_1 \cap S_2$ so $\vec{u} \in S_1$. Similarly $\vec{v} \in S_1 \cap S_2$ so $\vec{v} \in S_1$. Now because we know S_1 is a subspace, we know that it is closed under vector addition. Thus $\vec{u} + \vec{v} \in S_1$.

Step 2: Because $\vec{u} \in S_1 \cap S_2$ so $\vec{u} \in S_2$. Similarly $\vec{v} \in S_1 \cap S_2$ so $\vec{v} \in S_2$. Now because we know S_2 is a subspace, we know that it is closed under vector addition. Thus $\vec{u} + \vec{v} \in S_2$.

Step 3: Because $\vec{u} + \vec{v} \in S_1$ and also $\vec{u} + \vec{v} \in S_2$ we conclude that $\vec{u} + \vec{v} \in S_1 \cap S_2$.

Why is $S_1 \cap S_2$ closed under scalar multiplication?

Take a vector $\vec{v}$ from $S_1 \cap S_2$ and a scalar $c \in \mathbb{R}$.

We need to explain why $c\vec{v}$ is also in $S_1 \cap S_2$. We'll split that into three steps:

1. We'll explain that $c\vec{v} \in S_1$.
2. Then we'll explain that $c\vec{v} \in S_2$.
3. Thus $c\vec{v} \in S_1 \cap S_2$.

Step 1: Because $\vec{v} \in S_1 \cap S_2$ so $\vec{v} \in S_1$. Now because we know S_1 is a subspace, we know that it is closed under scalar multiplication. Thus $c\vec{v} \in S_1$.

Step 2: Because $\vec{v} \in S_1 \cap S_2$ so $\vec{v} \in S_2$. Now because we know S_2 is a subspace, we know that it is closed under scalar multiplication. Thus $c\vec{v} \in S_2$.

Step 3: Because $c\vec{v} \in S_1$ and also $c\vec{v} \in S_2$ we conclude that $c\vec{v} \in S_1 \cap S_2$.



Homogeneous systems

Video 9.2.7

Remember that a homogeneous system of linear equations is just a system where the constant terms are all zero.

For example:

$$\begin{cases} x_1 + x_2 + x_3 &= 0 \\ -x_1 + 7x_2 &= 0 \\ x_1 + 2x_2 + 3x_3 &= 0 \end{cases}.$$

The set of solutions is obviously a subset $S \subset \mathbb{R}^3$.
So we can ask: is it a subspace?

Yes! Homogeneous system $\Rightarrow$ Solutions form subspace.

Theorem (The solution set of a homogeneous linear system.)

Consider a homogeneous system S of linear equations in n variables.

Recall that the solution set of this system is a subset of $\mathbb{R}^n$.

In fact: the solution set will be a subspace.

This fact is important for two reasons:

- It gives us an easy way to check that certain examples of subsets are in fact subspaces.
- It connected this abstract topic (subspaces) directly to an important concrete example: sets of solutions of homogeneous linear systems.

Here's the example we did earlier:

Exercise

Consider the following subset of $\mathbb{R}^3$

$$S = \{(x_1, x_2, x_3) \in \mathbb{R}^3, x_1 + 2x_2 - 3x_3 = 0\} \rightarrow \text{homogeneous linear equation}$$

Is it a subspace of $\mathbb{R}^3$?

Prove your answer.

Proof

Note that S is exactly the set of solutions to a homogeneous linear system. Thus by the theorem, it is a subspace.

Proof of the theorem.

Consider a general homogeneous system:

$$\begin{cases} c_{11}x_1 + \cdots + c_{1n}x_n = 0 \\ \vdots \\ c_{k1}x_1 + \cdots + c_{kn}x_n = 0 \end{cases}$$

The clearest way to understand this system is to rewrite it as a matrix equation and exploit the algebra of matrices.

Remember that the n -tuple $(x_1, \dots, x_n)$ will solve this linear system, and hence lie in the solution set S , if and only if the corresponding matrix equation is true:

$$\begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & \vdots & \vdots \\ c_{k1} & \cdots & c_{kn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Proof of the theorem (continued).

So let's check the subspace axioms one-by-one:

Subspace Axiom #1:

Does $(0, \dots, 0) \in \mathbb{R}^n$ lie in the subset $S \subset \mathbb{R}^n$? In other words, is it a solution of the system?

It will be if the corresponding matrix equation is satisfied.

Let's check:

$$\begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & \vdots & \vdots \\ c_{k1} & \cdots & c_{kn} \end{bmatrix} \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}.$$

This is obviously true, so Axiom #1 holds.

Proof of the theorem (continued).

So let's check the subspace axioms one-by-one:

Subspace Axiom #2:

Take two vectors $(x_1, \dots, x_n), (y_1, \dots, y_n) \in S$.

In other words, these vectors are solutions of the system.

Is it also true that their sum

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n)$$

also lies in the set of solutions S ?

This will be true if the corresponding matrix equation is true.

Let's check it.

Proof of the theorem (continued).

$$\begin{aligned}
 & \begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & \vdots & \vdots \\ c_{k1} & \cdots & c_{kn} \end{bmatrix} \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix} \\
 &= \begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & \vdots & \vdots \\ c_{k1} & \cdots & c_{kn} \end{bmatrix} \left(\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \right) \quad \text{(Vector addition becomes sum of column vectors.)} \\
 &= \begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & \vdots & \vdots \\ c_{k1} & \cdots & c_{kn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & \vdots & \vdots \\ c_{k1} & \cdots & c_{kn} \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \quad \text{(Distributive law of matrix multiplication.)} \\
 &= \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \quad \text{(Because the two vectors are assumed to be solutions.)}
 \end{aligned}$$

Proof of the theorem (continued).

So let's check the subspace axioms one-by-one:

Subspace Axiom #3:

Take a vector $(x_1, \dots, x_n) \in S$.

In other words, this vector is a solution of the system.

Take a scalar $\lambda \in \mathbb{R}$.

Is it also true that the scalar multiple

$$\lambda(x_1, \dots, x_n) = (\lambda x_1, \dots, \lambda x_n)$$

also lies in the set of solutions S ?

This will be true if the corresponding matrix equation is true.

Let's check it.

Proof of the theorem (continued).

$$\begin{aligned}
 & \begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & \vdots & \vdots \\ c_{k1} & \cdots & c_{kn} \end{bmatrix} \begin{bmatrix} \lambda x_1 \\ \vdots \\ \vdots \\ \lambda x_n \end{bmatrix} \\
 &= \begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & \vdots & \vdots \\ c_{k1} & \cdots & c_{kn} \end{bmatrix} \left(\lambda \begin{bmatrix} x_1 \\ \vdots \\ \vdots \\ x_n \end{bmatrix} \right) \\
 &= \lambda \left(\begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & \vdots & \vdots \\ c_{k1} & \cdots & c_{kn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ \vdots \\ x_n \end{bmatrix} \right) \quad \text{(This is a basic} \\
 & \quad \text{rule of matrix} \\
 & \quad \text{multiplication.)} \\
 &= \lambda \begin{bmatrix} 0 \\ \vdots \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ \vdots \\ 0 \end{bmatrix} \quad \text{(Because the two vectors} \\
 & \quad \text{are assumed to be} \\
 & \quad \text{solutions.)}
 \end{aligned}$$

